HACKBOATS®

LEADERSHIP EDUCATION

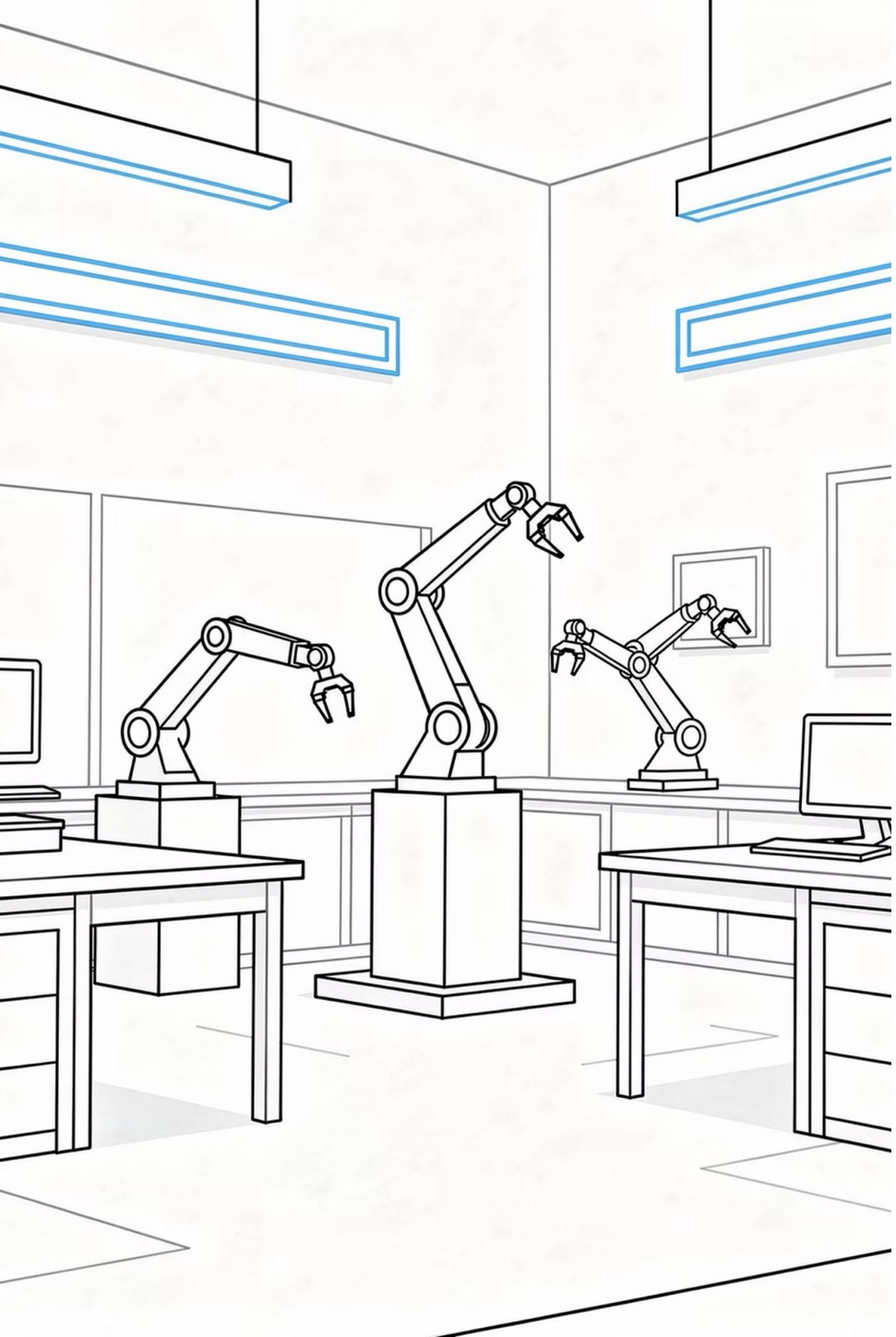
Drones & Space Science

Robotics & Artificial Intelligence

Animatronics & Automation

Elevate Your Leadership

hackboats.com



INDUSTRY–ACADEMIA STRATEGIC COLLABORATION

Center of Excellence (CoE) Robotics & Intelligent Systems

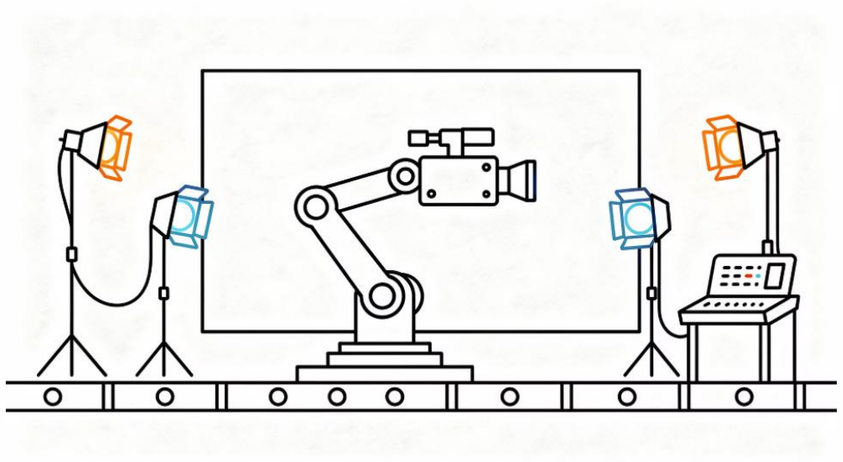
A transformative partnership to build India's next generation of industry-ready engineers — powered by hands-on robotics, AI, and automation education from Day One.

Presented by HackBoats · Robotics & Innovation Company

HackBoats – Industry Background

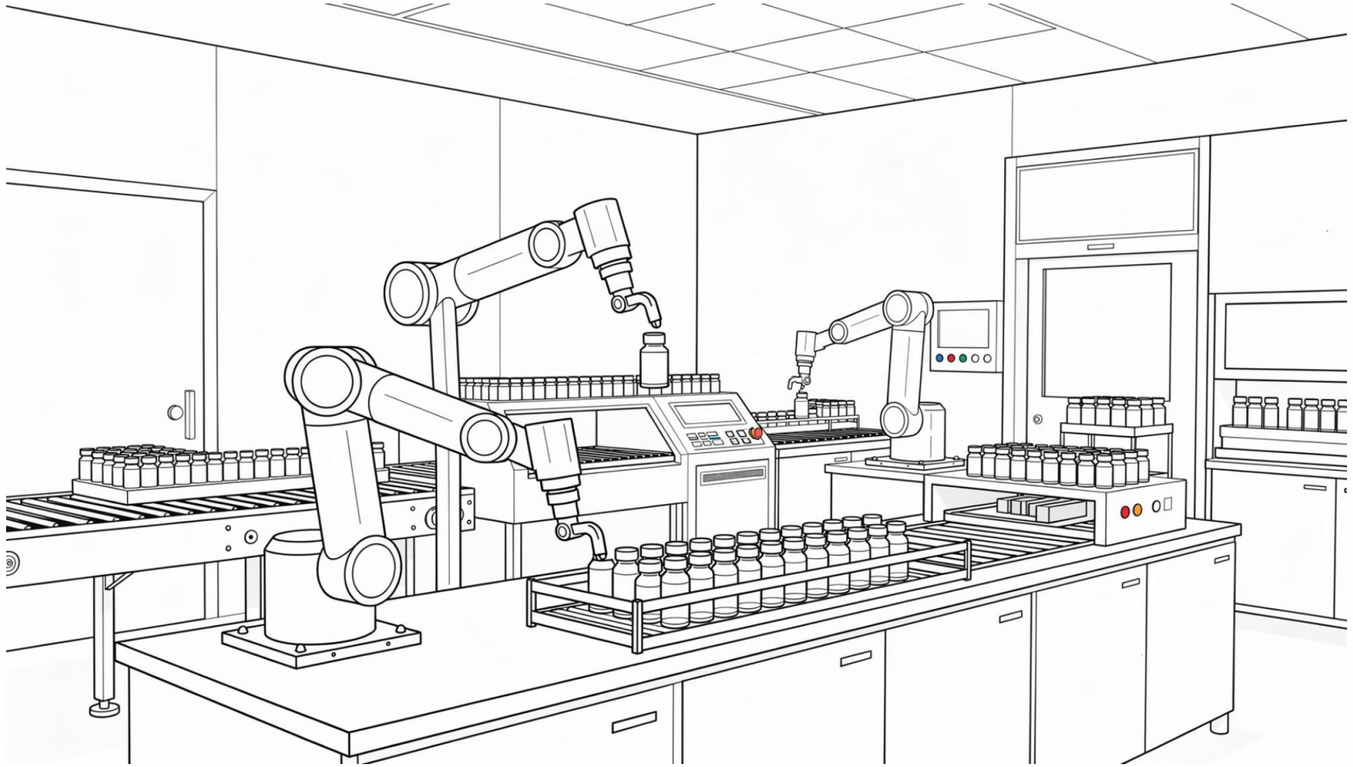
Film & Entertainment Technology

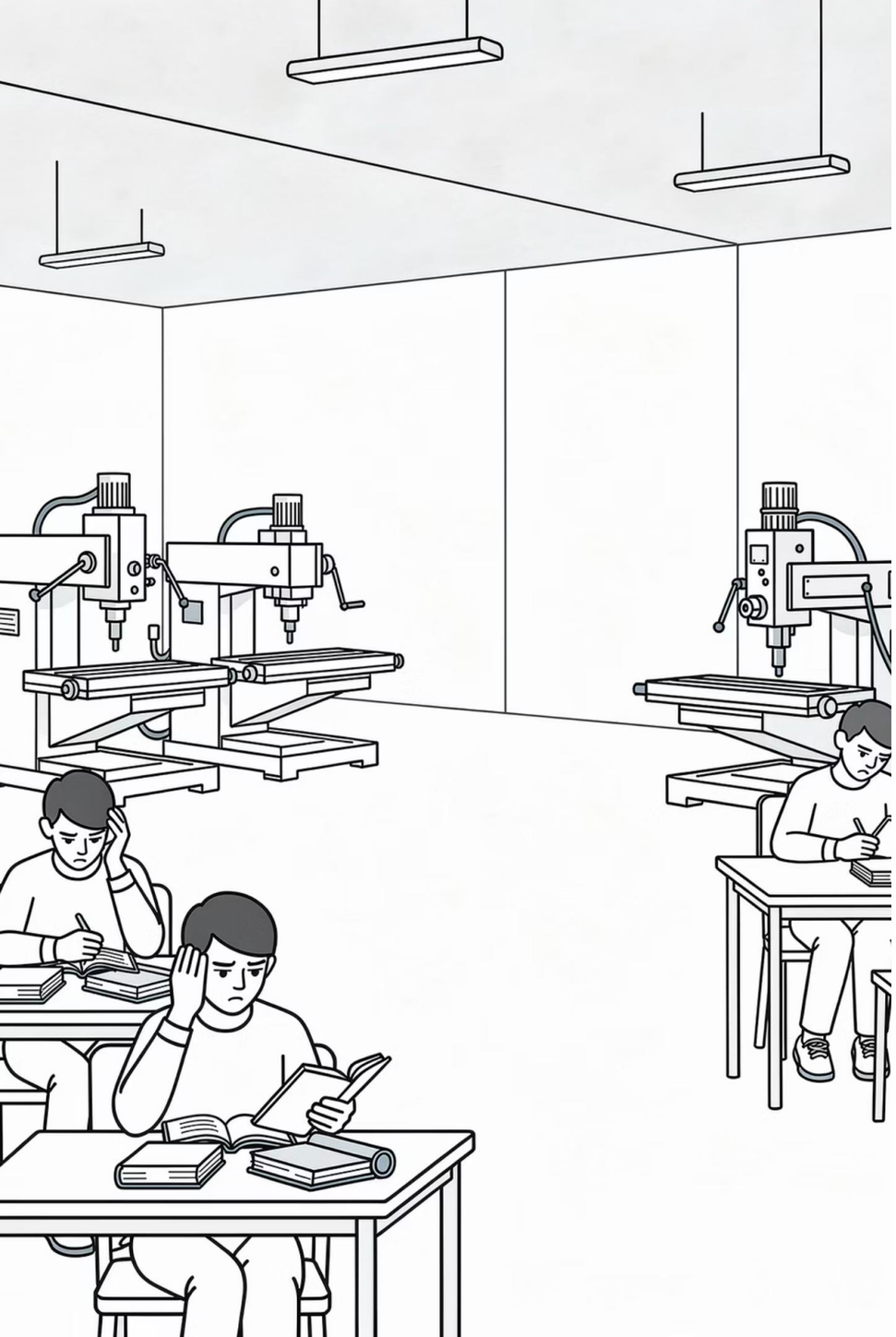
Robotics solutions for film production, special effects, camera automation, and advanced filming technology.



Pharmaceutical & Industrial Automation

Automation solutions, intelligent monitoring systems, and smart robotics technologies for pharmaceutical and industrial environments.





CHAPTER 1 THE CHALLENGE

The Industry Problem

India produces over **1.5 million engineers annually**, yet industry leaders consistently report that fewer than 20% are employable in core technology roles. The root cause lies not in academic rigor, but in the absence of practical, hands-on exposure to modern technologies like robotics, automation, and AI.

Skill Gap Crisis

Curriculum lags behind industry demands by 5–8 years, leaving graduates unprepared for modern workplaces.

No Hands-On Exposure

Most colleges lack robotics labs and automation infrastructure, limiting students to theory-only learning.

Employability Deficit

Core engineering companies struggle to find job-ready talent with practical project experience and technical portfolios.

CHAPTER 2 YOUR INDUSTRY PARTNER

About HackBoats

HackBoats is a Robotics Product Development and innovation company at the forefront of building industry-ready engineering talent across India. With deep expertise spanning robotics, drones, AI, automation, and embedded systems, HackBoats has partnered with engineering institutions to design and deploy hands-on innovation programs that bridge the gap between classroom learning and industry expectations.

Robotics & AI Expertise

End-to-end capabilities in designing, building, and deploying robotic systems and intelligent automation solutions.

Drone & UAV Technology

Specialized programs in drone design, assembly, flight operations, and real-world UAV applications across industries.

Embedded Systems & IoT

Deep proficiency in microcontroller programming, sensor integration, and Internet of Things ecosystems.

Proven Academic Partnerships

Track record of successful collaborations with engineering colleges across India, delivering innovation labs and structured learning programs. With over 25 Colleges

CHAPTER 3 THE VISION

Vision of the CoE

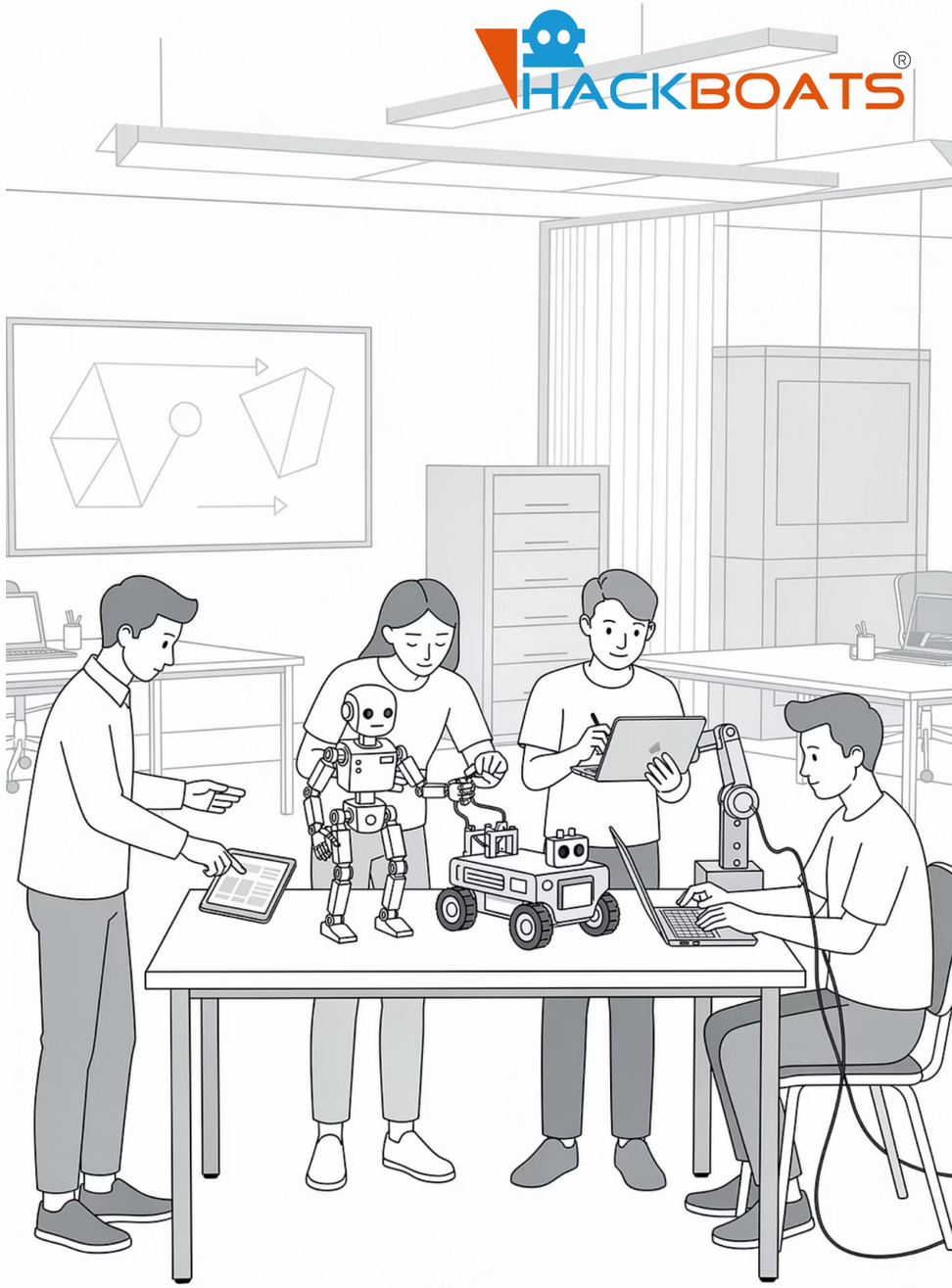
The Center of Excellence in Robotics & Intelligent Systems will establish a **permanent innovation ecosystem** within the college campus — transforming how engineering students learn, build, and prepare for their careers from the very first year.

What We Will Create

- A state-of-the-art Robotics & AI Lab on campus
- Industry-aligned curriculum with project-based learning
- A culture of innovation, prototyping, and entrepreneurship
- Direct industry connections from first year onward

Why It Matters

Students gain **3 full years** of hands-on exposure — building robots, programming AI, and working on real-world automation challenges. This isn't a workshop. It's a **transformation of engineering education** at your institution.



CHAPTER 4 INVESTMENT

Infrastructure Investment

HackBoats will invest **₹20,00,000** to design, equip, and deploy a fully operational Robotics & AI Lab inside your college campus. This investment covers everything from advanced hardware to innovation infrastructure — at **zero capital cost to the institution**.

₹20,00,000/-

Total Investment

Fully funded by HackBoats for lab setup and deployment

0

Cost to College

Zero capital expenditure required from the institution

Lab Components

1

Robotics Kits

Advanced robotic arms, mobile platforms, and programmable robot kits for hands-on building and experimentation.

2

Sensors & Microcontrollers

Arduino, Raspberry Pi, ESP32, and an extensive library of sensors — ultrasonic, IR, gyroscope, LiDAR, and more.

3

Tools & Equipment

3D printers, soldering stations, oscilloscopes, multimeters, and a complete prototyping workbench setup.

4

Innovation Infrastructure

Software licenses, AI/ML frameworks, drone assembly kits, and IoT development boards for cutting-edge projects.

Program Structure

The CoE operates on a carefully designed **3-year engagement model** that takes students from foundational skills to advanced industry-level project work — ensuring deep, compounding learning over their engineering journey.

Year 1 — Foundation

Core robotics fundamentals, basic programming, sensor integration, and introductory projects.

Year 2 — Specialization

Advanced domains: AI, drones, IoT, and automation. Students choose focus areas and build complex projects.

Year 3 — Industry Readiness

Capstone projects, industry internships, competition preparation, and technical portfolio development.

3

Years Duration

Full engagement from first year to third year

500

Students Per Year

First-year engineering intake enrolled annually

1,500

Total Students

Over the complete 3-year agreement period

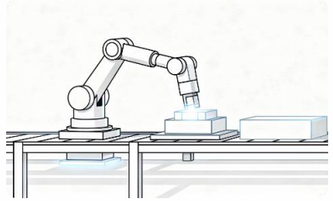
₹5K

Fee Per Student

For the entire 3-year program engagement

Technology Areas Covered

The CoE curriculum spans **five high-demand technology domains** — each representing a rapidly growing industry vertical where Indian engineering graduates are increasingly sought after by both startups and global enterprises.

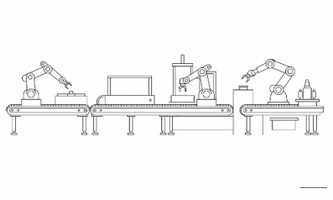


Robotics Systems

Design, build, and program robotic platforms — from mobile rovers to multi-axis manipulators for industrial applications.

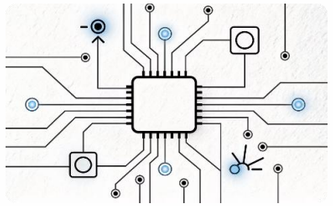
Drones & UAV Technology

Drone assembly, flight dynamics, autonomous navigation, and applications in agriculture, surveillance, and logistics.



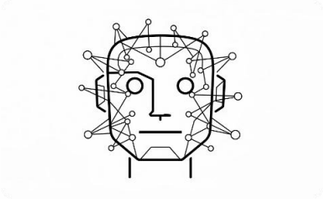
Industrial Automation

PLC programming, SCADA systems, conveyor automation, and smart manufacturing processes used in Industry 4.0.



Embedded Systems & IoT

Microcontroller programming, real-time systems, sensor networks, and connected IoT device development.

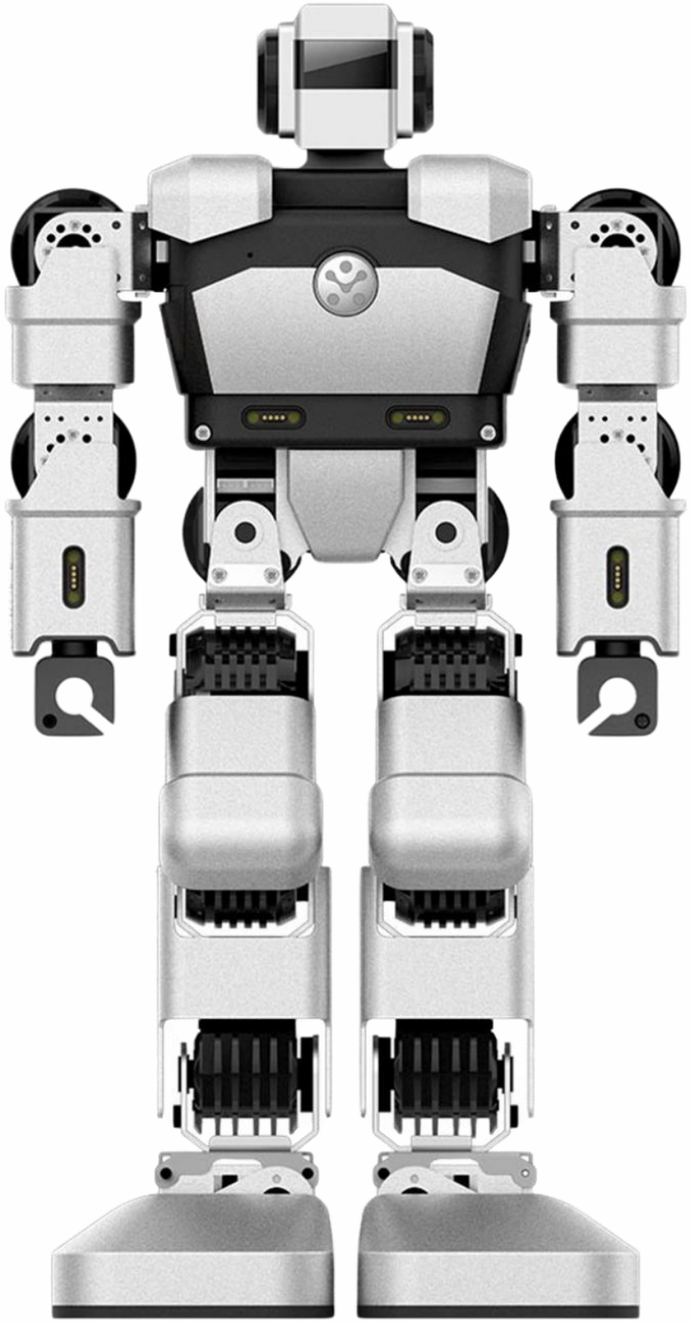


AI-Enabled Robotics

Machine learning for vision systems, path planning algorithms, and intelligent decision-making in autonomous robots.

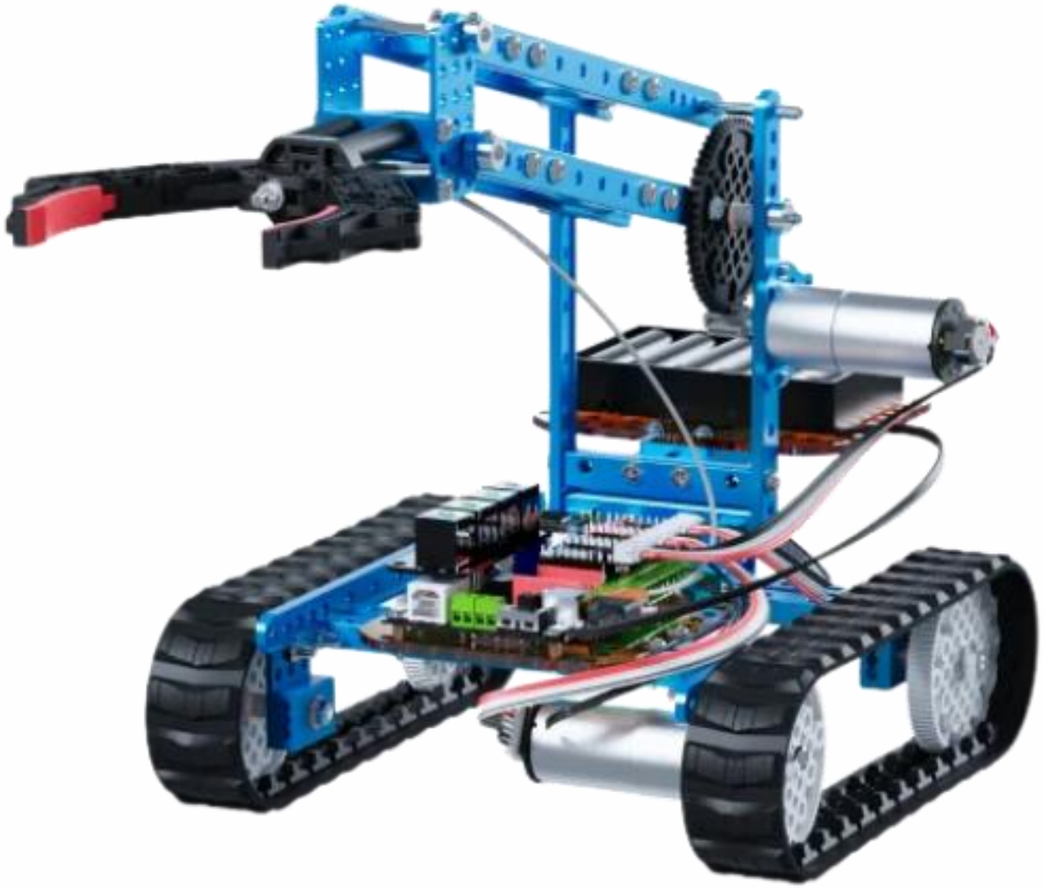
DOBOT
Magician E6

HACKBOATS®

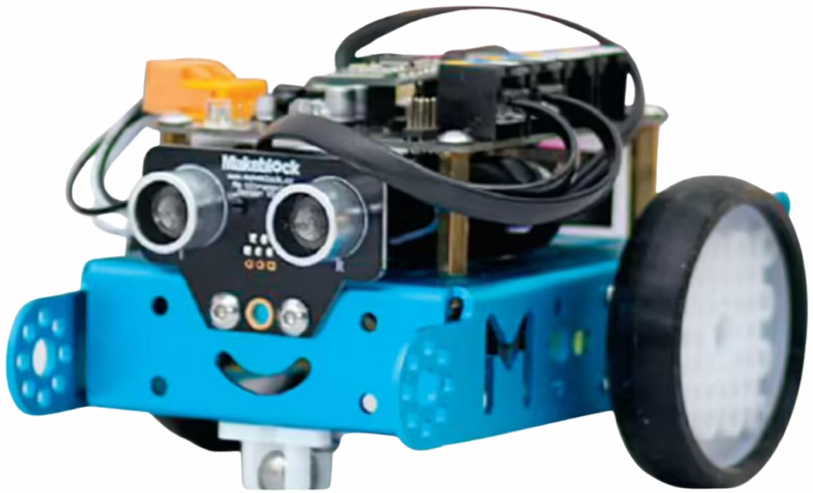


DOBOT
DOBOT Magician

UNITREE
02

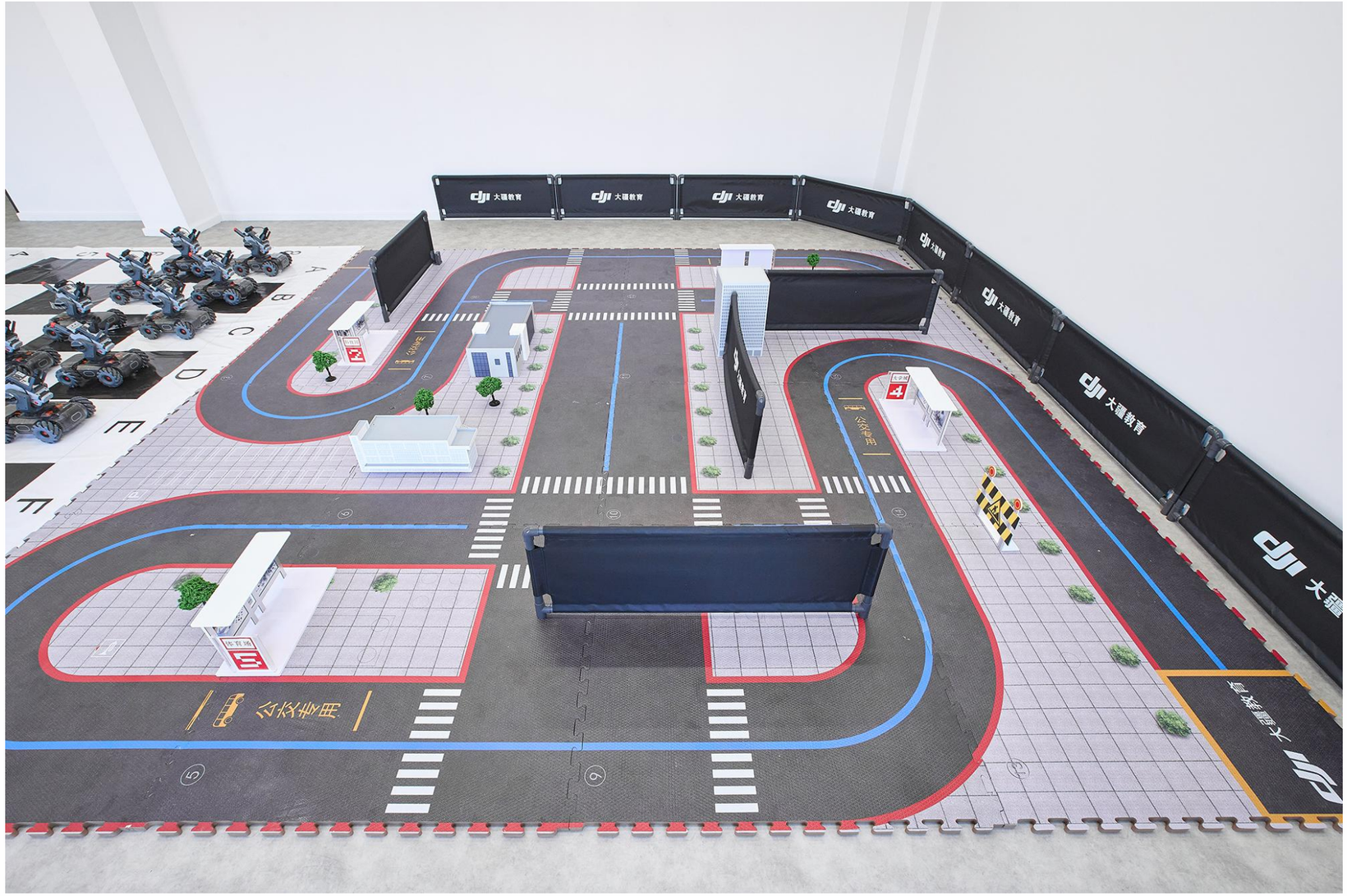


•**Controller Board:** MegaPi main controller
•**Processor:** ATMEGA2560-16AU (Arduino-compatible)
•**Clock Speed:** 16 MHz
•**Motor Driver:** Integrated high-power motor drivers
Control Platforms: PC, mobile app, tablet**Block-based programming (Scratch / mBlock)**
Arduino IDE (C/C++)
Node.js
Python (with Raspberry Pi integration)



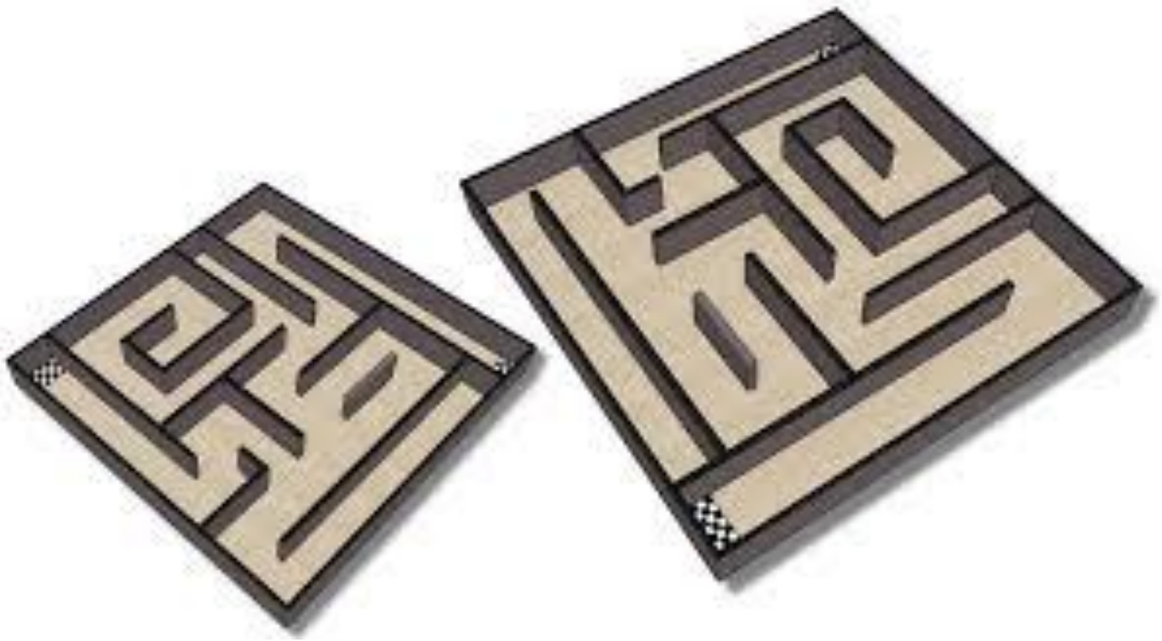
- **Main Controller:** CyberPi smart controller
- **Processor:** ESP32-WROVER-B dual-core processor
- **Flash Memory:** 8 MB
- **Display:** 1.44-inch full-color display
- **Speaker:** Built-in speaker
- **LEDs:** 5 RGB programmable LEDs
- **Battery:** 18650 rechargeable Li-ion battery (≈2500 mAh)
- **Battery Life:** Up to ~5 hours on a single charge

Smart City Arena



Soccer Arena

Maze Arena



Visiting Robot

Institutional Support Requirements

Basic infrastructure support required from the institution for successful implementation of the Robotics & Intelligent Systems CoE.

Infrastructure Support from College

Dedicated Lab Space

Approximately 2,000 sq.ft. of dedicated laboratory space for the CoE.

Interior Setup

Modern interior design suitable for a robotics and innovation lab environment.

Work Tables & Workstations

Student work tables and individual workstations for hands-on project work.

Air Conditioning

Climate-controlled laboratory environment for equipment and student comfort.

Lighting & Electrical

Adequate lighting, fans, and electrical connections throughout the lab.

Minimum 15 Desktop Computers

For programming, robotics simulation, and software development activities.

Internet Connectivity

High-speed internet for IoT, AI development, and cloud-based platforms.

> HackBoats will invest ₹20,00,000 to establish the Robotics & AI Lab and provide technical expertise, training, and industry engagement.

Student Outcomes

Graduates of the CoE program will not simply hold a degree — they will carry a **demonstrable portfolio** of real-world projects, practical skills, and an innovation mindset that sets them apart in the competitive engineering job market.

Hands-On Engineering Skills

Proficiency in building, programming, and debugging real robotic and automation systems — not just theory.

Innovation Mindset

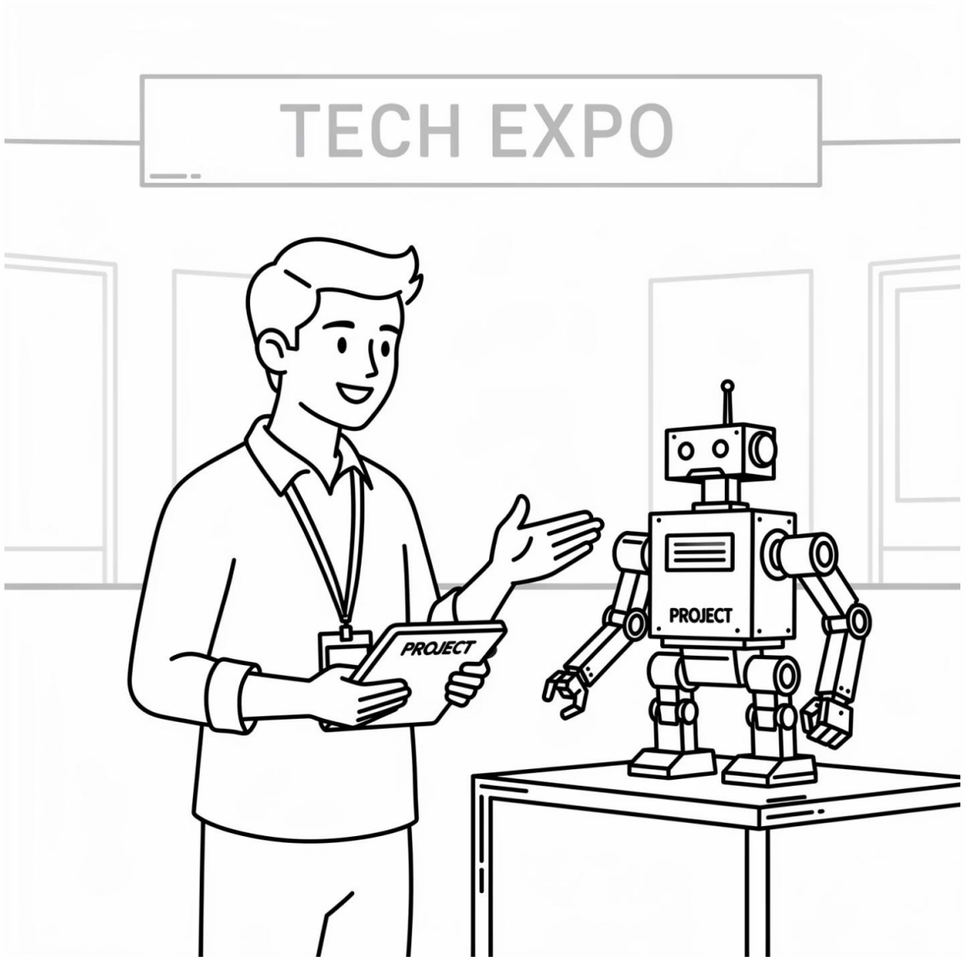
Problem-solving through design thinking, rapid prototyping, and iterative development — the startup way.

Industry-Grade Projects

Capstone and semester projects aligned with actual industry use cases in robotics, drones, and automation.

Technical Portfolio

A curated portfolio of projects, certifications, and competition entries that serves as a powerful differentiator during placement.



Placement & Industry Exposure

Beyond skill building, HackBoats brings its **industry network directly to your campus**. We will organize dedicated placement drives, connect students with hiring companies, and provide mentorship that accelerates career readiness across core engineering sectors.

Dedicated Placement Drives

Campus recruitment events with companies actively hiring in robotics, drones, automation, and embedded systems.

Industry Mentorship

One-on-one guidance from practicing engineers and industry professionals to prepare students for real-world roles.

Career Readiness

Resume building, technical interview preparation, and soft skills training tailored for core engineering careers.

Hiring Industries We Connect Students With

Robotics	Drone Tech
Automation	Core Engineering

LET'S BUILD TOGETHER

Partnership Proposal

We invite your institution to join hands with HackBoats in a **3-year strategic collaboration** to establish the Center of Excellence in Robotics & Intelligent Systems — a partnership that elevates your college's brand, enriches student outcomes, and positions your institution as a leader in innovation-driven engineering education.

Weeks 1–2

Agreement & Planning MoU signing, lab space identification, infrastructure planning, and curriculum alignment.

Weeks 5 Onward

Lab Setup & Deployment₹20L investment deployed — equipment installation, software setup, and lab inauguration.

1 2 3 4

Weeks 3–4

Program Launch First batch of 500 students onboarded, regular sessions begin, and innovation activities kick off.

Year 1–3

Full Operations Continuous program delivery, placement drives, competitions, and annual outcome reviews.

> **Ready to transform engineering education at your institution?**Let's start the conversation. HackBoats is prepared to deploy, operate, and deliver measurable impact — starting today.

Contact Us
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www.hackboats.com

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